

interface, is a competitive solution from TI, and is apt for applications like home security. Ganapathy says, "We have launched C3200 as a module and chipset. It consumes very low power and can operate on batteries. There are very few modules in the market that can operate on batteries."

Talking about a feature trend on the software side, Dhananjay Kulkarni, COO, Maven Systems Pvt Ltd, informs, "We now see mesh platforms being made available on more and more RF and Wi-Fi modules. Mesh algorithms allow a reach of several kilometres even if point-to-point range is only few tens or hundreds of metres."

He adds, "This is especially useful in case of applications such as street lighting, smartmeters, solar panels and windmill farms. The mesh-enabled RF and Wi-Fi modules convert many applications from wired to wireless and provide redundancy/high availability in the process."

General selection criteria

Listed below are parameters that you could consider before locking down on a particular module for your next design.

Select the right wireless technology. It is important to choose the right kind of wireless standards that the end application would support. Ganapathy feels, the designer needs to figure out the right technology such as NFC, Bluetooth, BLE, ZigBee, 6LoWPAN, Wi-Fi or even proprietary RF for the end application. Each of these come with their own share of advantages.

He says, "NFC uses very low power, whereas BLE requires more power. ZigBee or 6LoWPAN provide designers with an option of meshing, which is not available with other technologies. With meshing, even if one node fails on the network, it is possible to get data via other active nodes." He adds, "6LoWPAN and Internet protocol (IP) can help in easy connectivity to IP based networks, making connecting to the cloud or the Internet easy."

"Bluetooth has been at the forefront of connecting sensors, boards,

Pricing trends

"Prices vary from ₹ 1000 to ₹ 5000, depending on range, availability of the mesh algorithm, availability of gateway and cloud application."

—**Dhananjay Kulkarni, COO, Maven Systems Pvt Ltd**

"Prices have reduced slightly in the past few months. There seems to be no local-made Wi-Fi module products in India yet, with almost 100 per cent being imported from China. We have noticed the demand coming from large cities like Bengaluru, Mumbai, New Delhi and Hyderabad, and even from smaller towns such as Rajkot, Gwalior and Coimbatore. It shows that the IoT has caught the imagination of India's electronics research community at all levels, in colleges as well as in the industry."

—**Alok Jain, co-founder, InkOcean.in**

"The US\$ 5 to US\$ 6 category is becoming common for the IoT market. Selling modules at very low costs is not lucrative for manufacturers, as the investment that goes into making these products is huge. They are figuring out ways of including microcontrollers and software stacks into the module, which would significantly reduce the design time at system level. This helps in increasing the value of the modules."

—**Dhiraj Sogani, GM and Sr VP, Systems Business Unit, Redpine Signals, Inc.**

Point of view: Parameters to select the right module

Existing certifications. With more and more regulations on RF coming in, it is important to keep costs low and speed up the development as well as the launch of the product. Having modules already certified takes care of that.

Noise contribution and immunity. You would not want your module and, hence, the design/product to be a contributor to the overall system noise. You would also not want it to be fragile and too sensitive to external noise.

Additional development efforts. We select modules that could help us in getting to the market faster and cheaper.

Power consumption. Given that most devices going forward are going to be battery-powered, power consumption is one of the most critical factors for selection. Standby and sleep currents, number of power-down or sleep modes available and overall effect on energy consumption are strong factors while selecting Wi-Fi or RF modules.

Size and cost. These were always one of the many other selection criteria and that hold true now too. Wi-Fi and RF modules are way cheaper than before and would cost almost the same. It is only the functionality and application that will eventually decide what to use where and how.

—**T. Anand, managing director, Knewron**

devices, accessories and peripherals to the Internet via various control devices like modems, mobile phones, dongles and PCs," says Niraj Jha, CEO and MD, Fab.to.Lab.

He adds, "In its latest avatar, the research on Smart Bluetooth (previously BLE/BT 4.0), as named by the pioneers in the industry in this field, is primarily focused on improving power consumption without significant compromises in range." There are various other protocols like Zigbee, Z-wave and EnOcean, which are available readily in modules for prototyping and experimenting with wireless.

Highlighting a problem of choosing Wi-Fi for an IoT application, Hymel shares, "Unlike Bluetooth, which allows devices to discover each other, Wi-Fi requires the client device

to make a connection with Wi-Fi access points (AP) first. This works well from a computer or smartphone where you can enter Wi-Fi credentials, but most IoT devices do not contain an easy-to-use interface, such as a keyboard or touchscreen."

Several companies have been employing novel approaches to solve this problem. He adds, "Electric Imp (electricimp.com), for example, has users send their service set identifiers (SSIDs) and AP passwords optically to the Imp device by holding the Imp to a screen (smartphone or monitor) that flashes white and black for 0s and 1s."

Range. This is another important factor to consider. "Energy metering is a classic example of an application where a robust, long-range technology must be deployed," says Ganapathy.

Sogani says, "It is important to